

Forecasting and Risk Management of Equity Index Volatility:

Evidence from Implied Volatility, GARCH-Type Models and HAR-RV
Forecasts

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0.1 Data Cleaning and Preprocessing

I consider daily price data for the S&P 500 over the period from January 2, 2015 to May 4, 2026. The dataset contains a total of 2,850 observations.

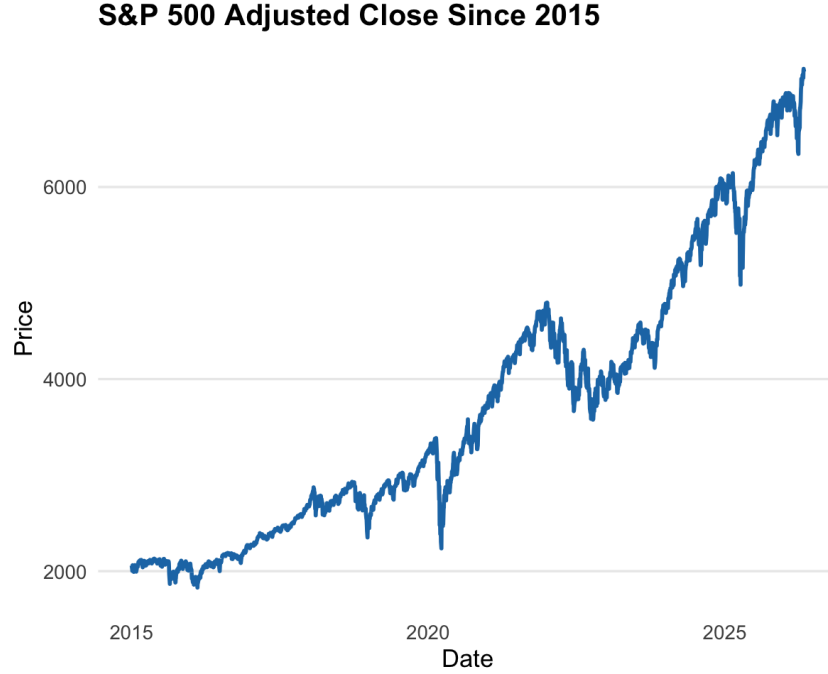


FIGURE 1 – Evolution of S&P 500 (Adjusted close price)

0.1.1 Data Construction

Let P_t denote the closing price at time t . I compute continuously compounded returns as :

$$r_t = \log(P_t) - \log(P_{t-1}) \quad (1)$$

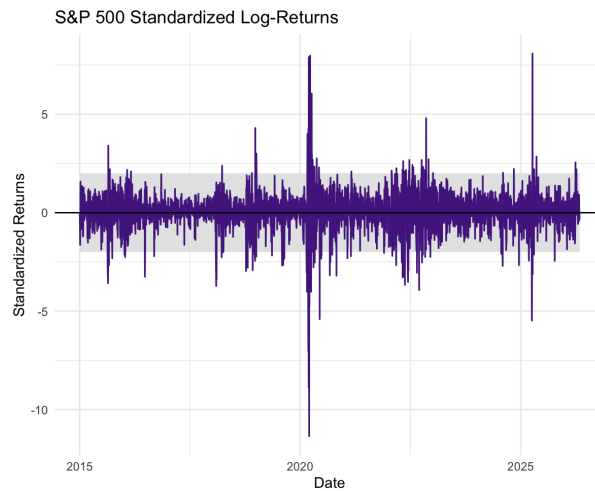


FIGURE 2 – log returns

This transformation ensures stationarity and is standard in financial econometrics. This stationarity is verified using ADF test :

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Augmented Dickey-Fuller Test

data: log_returns_std
Dickey-Fuller = -14.472, Lag order = 14, p-value = 0.01
alternative hypothesis: stationary

```

0.1.2 Data Cleaning

The dataset is carefully cleaned to ensure consistency and robustness :

- Missing values are removed.
- Duplicate timestamps are checked and eliminated.
- The time index is verified to ensure proper chronological ordering.

After cleaning, the dataset exhibits the following properties :

- Start date : January 2, 2015
- End date : May 4, 2026
- Number of observations : 2,850
- Missing values : 0
- Duplicate timestamps : 0

0.1.3 Outlier Treatment

Extreme returns are not removed, as they contain important information about market stress and volatility clustering. Instead, I flag extreme observations using a high quantile threshold :

$$|r_t| > q_{0.999} \quad (2)$$

where $q_{0.999}$ denotes the 99.9th percentile of absolute returns.

This approach preserves tail behavior while allowing us to monitor extreme events.

0.2 Realized Volatility Estimation

To measure time-varying market risk, I construct realized volatility using rolling windows of different lengths.

0.2.1 Definition

For a given window size w , realized volatility is defined as :

$$\sigma_t^{(w)} = \sqrt{252 \cdot \frac{1}{w} \sum_{i=1}^w r_{t-i}^2} \quad (3)$$

where :

- r_t denotes daily log-returns,
- w is the rolling window length,
- the factor 252 annualizes the volatility.

0.2.2 Rolling Windows

I compute realized volatility using multiple horizons :

- $w = 5$ days (weekly horizon),
- $w = 10$ days (bi-weekly horizon),
- $w = 21$ days (monthly horizon),
- $w = 63$ days (quarterly horizon).

This multi-scale approach allows us to capture both short-term and long-term volatility dynamics.

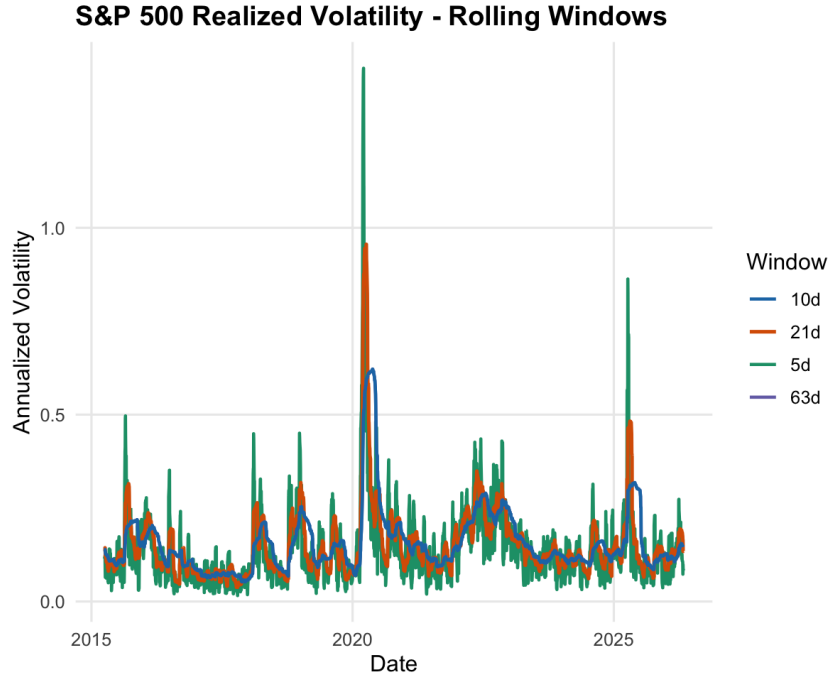


FIGURE 3 – Realized volatility computed for different time windows

0.2.3 Interpretation

Shorter windows react more quickly to market shocks but are noisier, while longer windows provide smoother estimates at the cost of slower responsiveness.

This trade-off is central in volatility modeling and will be further explored when evaluating forecasting performance.

0.2.4 Stylized Facts

Preliminary inspection of realized volatility reveals :

- Volatility clustering : periods of high volatility tend to persist.
- Time-varying variance : volatility is clearly not constant over time.
- Asymmetry : large negative returns are often associated with spikes in volatility.

These empirical features motivate the use of conditional heteroskedasticity models such as GARCH and its extensions.

0.3 Implied Volatility

0.3.1 Definition

Implied volatility represents the market's expectation of future volatility embedded in option prices. Unlike realized volatility, which is computed from historical returns, implied volatility is forward-looking and reflects investor sentiment and risk pricing.

In this study, I use the VIX index as a proxy for implied volatility. The VIX is constructed from S&P 500 index options and reflects the market's expectation of volatility over the next 30 days.

0.3.2 Data Construction

I retrieve daily observations of the VIX index over the same sample period as the S&P 500 data. Since the VIX is expressed in percentage terms, I convert it into decimal form :

$$IV_t = \frac{VIX_t}{100} \quad (4)$$

The implied volatility series is then aligned with the realized volatility dataset by matching trading dates and removing missing observations.

0.3.3 Implied vs Realized Volatility

A key objective of this project is to compare implied volatility with realized volatility. In particular, I focus on the 21-day realized volatility as a proxy for future volatility over a horizon comparable to the VIX.

Empirically, implied volatility tends to exceed realized volatility on average, reflecting a compensation demanded by investors for bearing volatility risk.

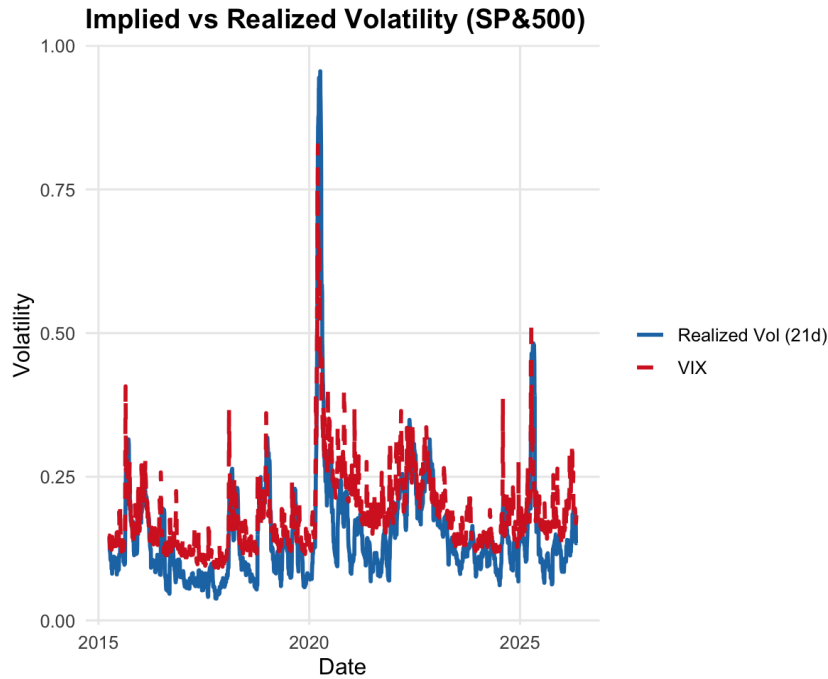


FIGURE 4 – Implied vs Realized volatility

0.3.4 Volatility Risk Premium

I define the volatility risk premium (VRP) as the difference between implied and realized volatility :

$$\text{VRP}_t = IV_t - \sigma_t^{(21)} \quad (5)$$

where $\sigma_t^{(21)}$ denotes the 21-day realized volatility.

The VRP captures the extent to which the market overestimates or underestimates future volatility :

- $\text{VRP}_t > 0$: implied volatility exceeds realized volatility, indicating a positive risk premium.
- $\text{VRP}_t < 0$: realized volatility exceeds implied volatility, often during periods of market stress.

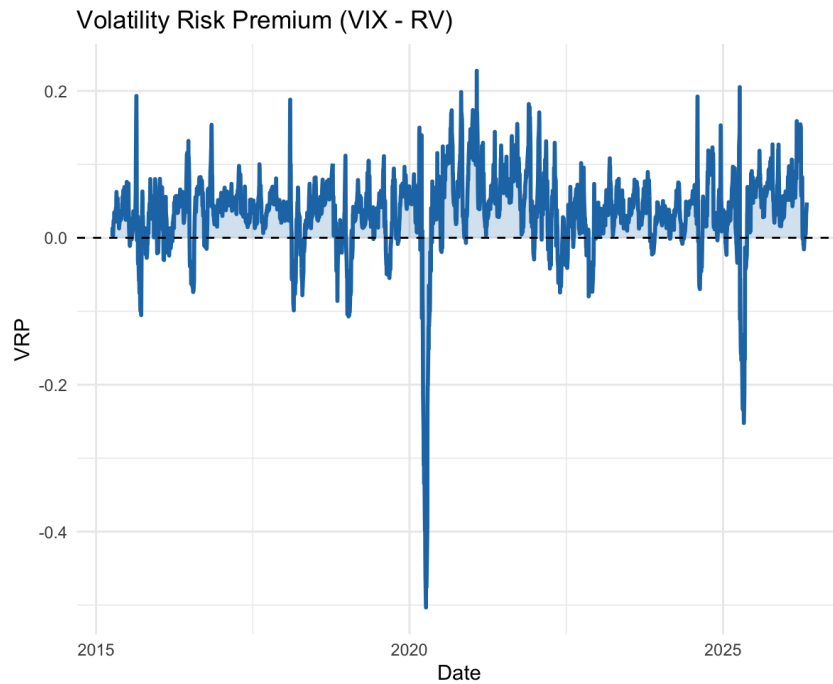


FIGURE 5 – Volatility risk premium

0.3.5 Economic Interpretation

The presence of a positive volatility risk premium is a well-documented phenomenon in financial markets. It reflects investors' aversion to volatility and their willingness to pay for protection against downside risk.

Analyzing the dynamics of the VRP provides insights into market expectations, risk perception, and potential mispricing, which are central to volatility modeling and forecasting.

0.3.6 Link with Volatility Modeling

Implied volatility serves as a natural benchmark for model-based forecasts. In the following sections, we will compare the predictive performance of GARCH-type models

against implied volatility to assess whether statistical models can outperform the market's expectations.

0.4 GARCH-Type Volatility Models

0.4.1 Motivation

Financial returns typically exhibit volatility clustering, meaning that periods of high volatility tend to be followed by periods of high volatility, and periods of low volatility tend to persist. This empirical regularity motivates the use of conditional heteroskedasticity models.

In this section, I estimate several GARCH-type models in order to forecast future volatility and compare model-based forecasts with realized and implied volatility benchmarks.

0.4.2 Baseline GARCH Model

The standard GARCH(1, 1) model specifies returns as :

$$r_t = \mu + \varepsilon_t \quad (6)$$

$$\varepsilon_t = \sigma_t z_t \quad (7)$$

where z_t is an i.i.d. motion process and σ_t^2 is the conditional variance.

The conditional variance follows :

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (8)$$

where :

- ω is the long-run variance component,
- α measures the short-term reaction to market shocks,
- β captures volatility persistence.

A high value of $\alpha + \beta$ indicates strong persistence in volatility.

0.4.3 Student-t Innovations

Since financial returns often exhibit fat tails, I also estimate GARCH models with Student-t motions rather than Gaussian motions. This allows the model to better capture extreme market movements.

0.4.4 EGARCH Model

To account for asymmetric volatility responses, I estimate an Exponential GARCH model. The EGARCH(1, 1) specification models the logarithm of conditional variance :

$$\log(\sigma_t^2) = \omega + \beta \log(\sigma_{t-1}^2) + \alpha (|z_{t-1}| - \mathbb{E}[|z_{t-1}|]) + \gamma z_{t-1} \quad (9)$$

The parameter γ captures the leverage effect. A negative value indicates that negative returns increase future volatility more than positive returns of the same magnitude.

0.4.5 GJR-GARCH Model

I also estimate a GJR-GARCH model, which introduces an explicit asymmetry term :

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \mathbb{I}_{\{\varepsilon_{t-1} < 0\}} \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2 \quad (10)$$

where $\mathbb{I}_{\{\varepsilon_{t-1} < 0\}}$ is an indicator function equal to one when the previous shock is negative.

If $\gamma > 0$, negative shocks have a larger impact on conditional volatility than positive shocks.

0.4.6 HAR-RV Model

In addition to GARCH-type models, I estimate a Heterogeneous Autoregressive Realized Volatility model. The HAR-RV model is a standard benchmark in realized volatility forecasting because it is simple, interpretable and highly competitive empirically.

The model is defined as :

$$RV_{t+1} = \beta_0 + \beta_d RV_t + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1} \quad (11)$$

where RV_t is the daily realized volatility, while $RV_t^{(w)}$ and $RV_t^{(m)}$ are weekly and monthly realized volatility components :

$$RV_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 RV_{t-i} \quad (12)$$

$$RV_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} RV_{t-i} \quad (13)$$

The HAR-RV model captures volatility persistence across different time horizons. Economically, this reflects the idea that market participants operate at different frequencies : short-term traders react to daily information, while institutional investors and longer-term agents influence volatility over weekly or monthly horizons.

Unlike GARCH models, which model conditional variance from return innovations, HAR-RV directly models realized volatility. This makes it a useful competing benchmark against both implied volatility and GARCH-family forecasts.

0.4.7 Model Specifications

I estimate the following models :

- GARCH(1, 1) with Student-t innovations,
- EGARCH(1, 1) with Student-t innovations,
- GJR-GARCH(1, 1) with Student-t innovations,
- HAR-RV model based on daily, weekly and monthly realized volatility components.

This model set allows us to compare implied volatility, parametric conditional volatility models and realized-volatility-based forecasting models.

0.4.8 Training and Testing Framework

The sample is split into an in-sample training period and an out-of-sample testing period. Models are initially estimated on the training sample ending on December 31, 2021. Forecasting performance is then evaluated on the out-of-sample period starting in January 2022.

To avoid look-ahead bias, GARCH-family forecasts are generated using a rolling estimation procedure. The models are re-estimated every 21 trading days using a moving window.

The HAR-RV model is estimated on the same training period and evaluated over the same out-of-sample testing period. Its predictors are constructed only from lagged realized volatility components, ensuring that no future information is used in the forecasting procedure.

0.4.9 Volatility Forecasting

For each model, I generate one-step-ahead volatility forecasts. Since the GARCH-family models are estimated on daily returns, the resulting daily volatility forecasts are annualized as :

$$\hat{\sigma}_{t,\text{ann}} = \sqrt{252} \cdot \hat{\sigma}_{t,\text{daily}} \quad (14)$$

For the HAR-RV specification, the forecasts are directly obtained from the realized volatility regression framework using lagged daily, weekly and monthly realized volatility components.

The forecasts are then compared with :

- 21-day realized volatility,
- VIX implied volatility,
- alternative GARCH-type model forecasts,
- HAR-RV realized volatility forecasts.

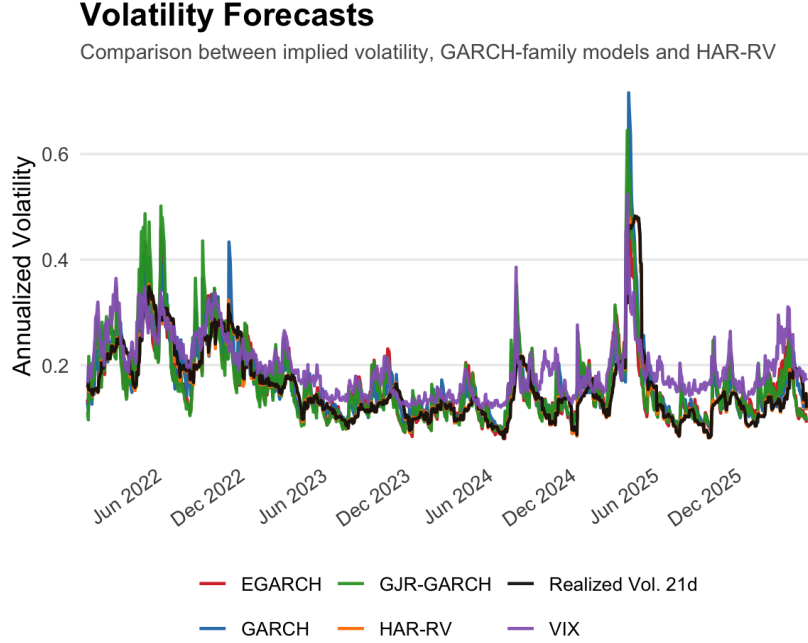


FIGURE 6 – Volatility forecasts comparison : VIX, GARCH-family models and HAR-RV

The figure highlights several important stylized facts. First, all volatility models successfully capture volatility clustering and periods of market stress, particularly during crisis episodes. Second, the HAR-RV model produces smoother forecasts due to its dependence on realized volatility persistence across multiple horizons. Finally, the GARCH-family models react more rapidly to abrupt market shocks because their dynamics are directly driven by return innovations.

0.4.10 Economic Interpretation

The purpose of this modeling framework is not only to identify the model with the best statistical fit, but also to assess whether econometric volatility forecasts can compete with the market-implied benchmark.

More importantly, the project compares three fundamentally different approaches to volatility modeling :

- market-implied volatility through the VIX,
- conditional volatility models through GARCH-family specifications,
- realized-volatility forecasting through the HAR-RV framework.

This comparison is particularly relevant in quantitative finance because each framework reflects a different view of volatility dynamics :

- the VIX reflects forward-looking market expectations,
- GARCH models capture conditional heteroskedasticity from return innovations,
- HAR-RV captures long-memory persistence in realized volatility.

The comparison between these competing approaches directly addresses the central research question of this project :

Can econometric and realized-volatility-based models outperform market-implied volatility expectations ?

0.5 Forecast Evaluation

0.5.1 Evaluation Framework

To assess the predictive performance of volatility models, I compare model-based forecasts with realized volatility using several loss functions :

- Mean Squared Error (MSE),
- Mean Absolute Error (MAE),
- QLIKE loss function.

The QLIKE loss is particularly relevant in volatility forecasting, as it is robust to measurement error in realized volatility and directly evaluates variance forecasts.

The evaluation framework includes :

- implied volatility forecasts (VIX),
- GARCH-family volatility forecasts,
- HAR-RV realized volatility forecasts.

This provides a broad comparison between market-implied, conditional-volatility and realized-volatility-based forecasting approaches.

0.5.2 Empirical Results

Table 1 reports the out-of-sample forecasting performance of the competing volatility models.

Model	MSE	MAE	QLIKE	Relative QLIKE
GARCH (Student-t)	0.00139	0.02475	-2.77466	1.049
EGARCH (Student-t)	0.00243	0.03533	-2.70404	1.022
GJR-GARCH (Student-t)	0.00271	0.03633	-2.70216	1.022
HAR-RV	0.00001	0.00217	-2.84045	1.074
VIX	0.00360	0.04790	-2.64506	1.000

TABLE 1 – Out-of-sample volatility forecast evaluation

The empirical results show that all econometric volatility models outperform the VIX benchmark across the different forecasting metrics.

Among the competing models, the HAR-RV specification achieves the strongest forecasting performance by a substantial margin. In particular, HAR-RV produces dramatically lower MSE and MAE values, suggesting that realized-volatility persistence contains significant predictive information for future volatility dynamics.

The GARCH Student-t model remains the strongest performer among the parametric conditional volatility models, outperforming both EGARCH and GJR-GARCH specifications.

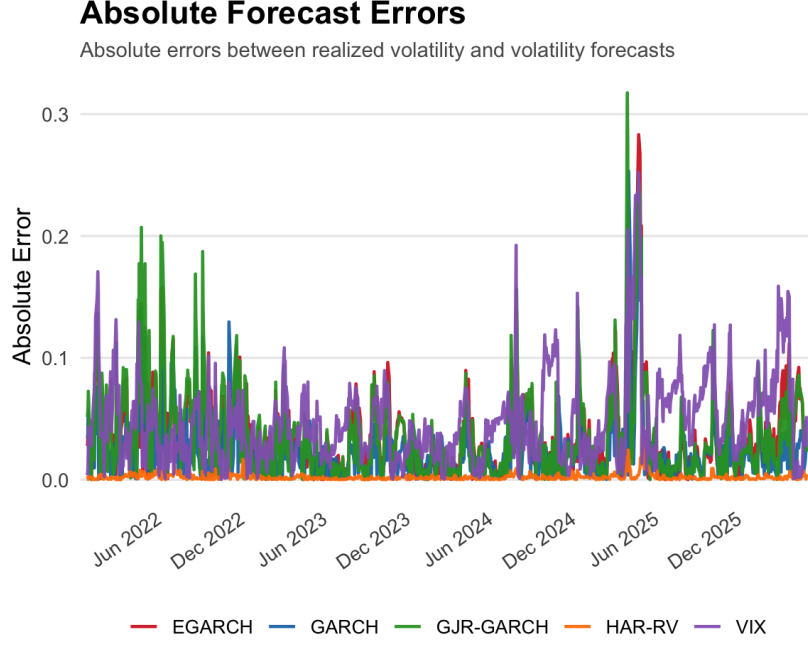


FIGURE 7 – Absolute forecast errors across volatility forecasting models

The forecast error dynamics further confirm the superior stability of the HAR-RV model. While GARCH-family forecasts react rapidly to market shocks, HAR-RV forecasts appear smoother and less sensitive to short-term noise, reflecting the long-memory structure embedded in realized volatility.

0.5.3 Diebold-Mariano Statistical Tests

While standard forecasting metrics such as MSE, MAE and QLIKE provide a ranking of model performance, they do not indicate whether the observed differences are statistically significant. To formally compare competing volatility forecasts, Diebold-Mariano (DM) tests were implemented.

The Diebold-Mariano framework evaluates the null hypothesis of equal predictive accuracy between two forecasting models. Let :

$$d_t = L(e_{1,t}) - L(e_{2,t})$$

where :

- $L(\cdot)$ denotes a forecasting loss function,
- $e_{1,t}$ and $e_{2,t}$ are the forecast errors of the two competing models.

Under the null hypothesis :

$$H_0 : \mathbb{E}[d_t] = 0$$

meaning that both models have equal predictive ability.

The DM statistic is asymptotically normally distributed :

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}}$$

where $\bar{d}$ is the average loss differential.

The tests were conducted using the MSE, MAE and QLIKE loss functions for the following comparisons :

- VIX vs GARCH Student-t,
- VIX vs EGARCH Student-t,
- GARCH Student-t vs EGARCH Student-t,
- HAR-RV vs VIX,
- HAR-RV vs GARCH Student-t,
- HAR-RV vs EGARCH Student-t,
- HAR-RV vs GJR-GARCH Student-t.

TABLE 2 – Diebold-Mariano Test Results

Model 1	Model 2	Loss	DM Statistic	p-value	Preferred Model
VIX	GARCH	MSE	12.81	< 0.001	GARCH
VIX	GARCH	MAE	20.88	< 0.001	GARCH
VIX	GARCH	QLIKE	19.54	< 0.001	GARCH
VIX	EGARCH	MSE	7.81	< 0.001	EGARCH
VIX	EGARCH	MAE	11.07	< 0.001	EGARCH
VIX	EGARCH	QLIKE	6.93	< 0.001	EGARCH
GARCH	EGARCH	MSE	-8.16	< 0.001	GARCH
GARCH	EGARCH	MAE	-14.43	< 0.001	GARCH
GARCH	EGARCH	QLIKE	-12.55	< 0.001	GARCH
HAR-RV	VIX	MSE	-18.67	< 0.001	HAR-RV
HAR-RV	VIX	MAE	-41.98	< 0.001	HAR-RV
HAR-RV	VIX	QLIKE	-27.70	< 0.001	HAR-RV
HAR-RV	GARCH	MSE	-9.80	< 0.001	HAR-RV
HAR-RV	GARCH	MAE	-27.59	< 0.001	HAR-RV
HAR-RV	GARCH	QLIKE	-18.64	< 0.001	HAR-RV
HAR-RV	EGARCH	MSE	-12.97	< 0.001	HAR-RV
HAR-RV	EGARCH	MAE	-32.49	< 0.001	HAR-RV
HAR-RV	EGARCH	QLIKE	-17.90	< 0.001	HAR-RV
HAR-RV	GJR-GARCH	MSE	-13.06	< 0.001	HAR-RV
HAR-RV	GJR-GARCH	MAE	-30.97	< 0.001	HAR-RV
HAR-RV	GJR-GARCH	QLIKE	-22.20	< 0.001	HAR-RV

The Diebold-Mariano results reveal several important findings.

First, both GARCH and EGARCH significantly outperform the VIX benchmark across all loss functions. This confirms that econometric volatility models contain predictive information beyond market-implied volatility alone.

Second, the GARCH Student-t specification significantly outperforms the EGARCH model under MSE, MAE and QLIKE losses. Despite its simpler symmetric structure, the standard GARCH model delivers stronger volatility forecasting performance than more complex asymmetric specifications in this empirical setting.

Most importantly, the HAR-RV model significantly outperforms all competing models, including the VIX and all GARCH-family specifications, across every forecasting metric considered in this study. The corresponding Diebold-Mariano statistics are extremely large in absolute value and associated with near-zero p-values, indicating highly robust

statistical significance.

These findings suggest that realized-volatility persistence contains substantially more predictive information about future volatility dynamics than both implied volatility and conditional variance models estimated from returns alone.

Economically, this result supports the heterogeneous market hypothesis underlying the HAR-RV framework. Market participants operating at different investment horizons generate long-memory dynamics in realized volatility, which are effectively captured by the HAR-RV specification.

Overall, the empirical evidence indicates that realized-volatility-based models provide the strongest forecasting performance in this dataset.

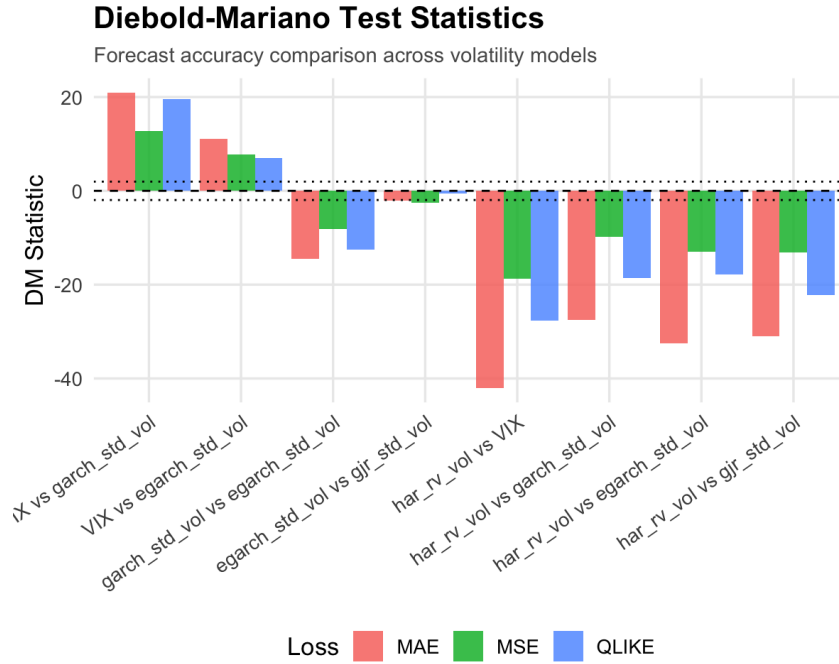


FIGURE 8 – Diebold-Mariano test statistics across volatility forecasting models

0.5.4 Discussion

Several important conclusions emerge from the empirical analysis.

First, all econometric volatility models outperform the VIX benchmark in terms of MSE, MAE and QLIKE loss functions. The Diebold-Mariano tests further confirm that these improvements are statistically significant.

Second, among the GARCH-family specifications, the standard GARCH(1,1) model with Student-t innovations delivers the best overall forecasting performance. This result highlights the importance of modeling heavy-tailed return distributions when forecasting financial market volatility.

Third, although EGARCH and GJR-GARCH explicitly incorporate leverage effects and asymmetric volatility dynamics, these extensions do not systematically improve out-of-sample volatility forecasting accuracy in this dataset.

Most importantly, the HAR-RV model strongly dominates all competing specifications. The HAR-RV forecasts exhibit significantly lower forecasting errors than implied volatility and GARCH-family models, suggesting that realized-volatility persistence contains substantial predictive information regarding future market volatility.

This result highlights a fundamental distinction between competing volatility modeling frameworks :

- implied volatility models incorporate forward-looking market expectations,
- GARCH-family models capture conditional heteroskedasticity from returns,
- HAR-RV captures long-memory persistence in realized volatility.

The empirical results suggest that realized-volatility-based models provide the strongest forecasting performance for this dataset.

However, the risk-management analysis reveals a more nuanced picture. Although HAR-RV dominates in terms of forecasting accuracy, the EGARCH specification performs better in the Value-at-Risk backtesting framework, particularly at the 5% level. This suggests that models optimized for volatility forecasting are not necessarily optimal for tail-risk estimation.

Overall, the results demonstrate that model selection in quantitative finance critically depends on the final objective :

- volatility forecasting,
- derivative pricing,
- portfolio allocation,
- or market risk management.

0.5.5 Economic Interpretation

The comparison between implied volatility, GARCH-family models and HAR-RV forecasts provides important insights into volatility dynamics and market efficiency.

The superior performance of the HAR-RV model suggests that realized volatility exhibits strong persistence across multiple investment horizons. This persistence appears to contain more predictive information than both option-implied volatility and conditional variance models estimated solely from returns.

At the same time, the strong performance of the VIX in the risk-management framework highlights the importance of forward-looking information embedded in option markets.

Overall, the empirical evidence suggests that volatility dynamics are driven by a combination of :

- market expectations,
- return innovations,
- and long-memory persistence in realized volatility.

This explains why combining realized-volatility measures, implied volatility and econometric volatility models remains central in modern quantitative finance.

0.5.6 Key Takeaway

Overall, the empirical results indicate that :

- all econometric volatility models significantly outperform the VIX benchmark,
- the GARCH Student-t specification is the strongest performer among the GARCH-family models,
- the HAR-RV model provides the best overall forecasting performance,
- EGARCH delivers the strongest tail-risk forecasting performance in the VaR backtesting framework.

These findings illustrate that realized-volatility-based models are highly competitive for volatility forecasting, while asymmetric GARCH models remain particularly relevant for market risk management applications.

0.6 Risk Management and Tail-Risk Backtesting

0.6.1 Value-at-Risk Framework

While volatility forecasting is a central component of quantitative finance, risk management applications require evaluating whether volatility models are capable of accurately capturing extreme downside events. For this reason, the rolling GARCH forecasts were extended to a Value-at-Risk (VaR) framework.

For a confidence level α , the one-day ahead VaR corresponds to the loss threshold that should only be exceeded with probability α :

$$VaR_{\alpha,t+1} = \mu_{t+1} + \sigma_{t+1}q_{\alpha}$$

where :

- μ_{t+1} is the conditional mean forecast,
- σ_{t+1} is the conditional volatility forecast,
- q_{α} is the quantile of the assumed innovation distribution.

The analysis focuses on the 1% and 5% VaR levels using the rolling forecasts generated by the GARCH, EGARCH and GJR-GARCH models with Student-t innovations.

0.6.2 Kupiec Unconditional Coverage Test

To evaluate whether the VaR forecasts produce the correct number of violations, the Kupiec unconditional coverage test was implemented.

Let :

$$\hat{p} = \frac{x}{N}$$

where :

- x is the number of observed violations,
- N is the total number of forecasts.

The Kupiec likelihood-ratio statistic is defined as :

$$LR_{uc} = -2 \log \left(\frac{(1 - \alpha)^{N-x} \alpha^x}{(1 - \hat{p})^{N-x} \hat{p}^x} \right)$$

Under the null hypothesis of correct unconditional coverage, the statistic follows a $\chi^2(1)$ distribution.

A rejection of the null hypothesis indicates that the model underestimates or overestimates downside risk.

0.6.3 Christoffersen Independence Test

In addition to the number of violations, a reliable VaR model should also produce violations that are independently distributed over time.

The Christoffersen independence test evaluates whether violations occur randomly or cluster during periods of market stress.

The test compares :

- a restricted model assuming independent violations,
- an unrestricted first-order Markov process.

A rejection of the null hypothesis would indicate volatility clustering not properly captured by the model.

0.6.4 Expected Shortfall Analysis

Beyond VaR exceedances, the analysis also considers Expected Shortfall (ES), which measures the average realized loss conditional on a VaR violation :

$$ES_{\alpha} = \mathbb{E} [r_t \mid r_t < VaR_{\alpha,t}]$$

Expected Shortfall provides additional information regarding the severity of tail losses and complements the VaR backtesting framework.

0.6.5 Empirical Backtesting Results

Table 2 reports the empirical VaR backtesting results for the different GARCH-family specifications.

TABLE 3 – VaR Backtesting Results

Model	α	Violations	Violation Rate	Kupiec p-value	Christoffersen p-value	ES
GARCH	1%	16	1.47%	0.148	0.230	-2.696
GARCH	5%	85	7.79%	0.0001	0.237	-2.014
EGARCH	1%	16	1.47%	0.148	0.230	-2.676
EGARCH	5%	72	6.60%	0.021	0.704	-2.078
GJR-GARCH	1%	16	1.47%	0.148	0.490	-2.398
GJR-GARCH	5%	79	7.24%	0.001	0.414	-2.028

The empirical results show that all models provide statistically acceptable VaR forecasts at the 1% level, as the Kupiec and Christoffersen tests are not rejected.

However, at the 5% level, all models fail the Kupiec test, indicating that the observed number of violations exceeds the theoretical level implied by the models. This suggests that downside risk is underestimated during moderately stressed market conditions.

Among the competing specifications, the EGARCH model delivers the best overall performance at the 5% level, with the lowest violation rate and the highest Kupiec p-value.

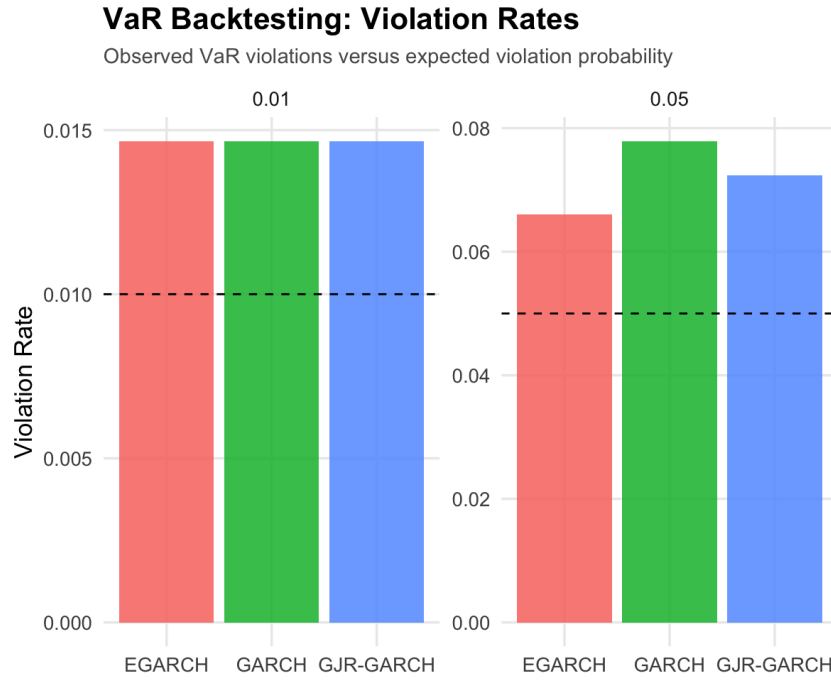


FIGURE 9 – Graph of violation rates for VAR 1% and 5%

0.6.6 Economic Interpretation

These results highlight an important distinction between volatility forecasting accuracy and tail-risk forecasting performance.

Although all GARCH-family models produce relatively accurate volatility forecasts, their ability to capture extreme downside events differs significantly. The superior performance of the EGARCH specification suggests that accounting for asymmetric volatility dynamics improves tail-risk estimation.

This result is economically consistent with the leverage effect observed in equity markets, where negative returns generate larger increases in future volatility than positive returns of the same magnitude.

Overall, the backtesting framework demonstrates that volatility models should not only be evaluated through forecasting loss functions such as MSE or QLIKE, but also through their capacity to capture market tail risk in a statistically robust manner.

0.7 Conclusion

This study investigates the ability of econometric models to forecast equity market volatility and compares their performance with market-implied expectations.

Using daily data on the S&P500, I construct realized volatility measures and evaluate several GARCH-type models, including GARCH, EGARCH, and GJR-GARCH specifications with Student-t innovations.

The empirical results show that :

- GARCH-type models outperform the VIX in terms of traditional error metrics such as MSE and MAE,
- accounting for fat tails via Student-t distributions significantly improves forecasting performance,

- more complex asymmetric models (EGARCH, GJR-GARCH) do not systematically outperform the standard GARCH specification.

However, when evaluated using the QLIKE loss function, implied volatility remains a strong benchmark, highlighting the difficulty of outperforming market-based forecasts.

Overall, the results suggest that while econometric models capture historical volatility dynamics effectively, they do not fully incorporate forward-looking information embedded in option prices.

This reinforces the view that combining statistical models with market-implied information is essential for accurate volatility forecasting.

0.8 Limitations and Extensions

While the results provide valuable insights, several limitations should be acknowledged.

0.8.1 Limitations

- The analysis relies on daily data, which may not fully capture high-frequency volatility dynamics.
- Realized volatility is approximated using rolling windows rather than high-frequency measures.
- The models are purely backward-looking and do not incorporate exogenous information.
- Only one asset is considered, limiting cross-sectional generalization.

0.8.2 Potential Extensions

Several extensions could significantly enhance the analysis.

High-Frequency Volatility Measures Using intraday data, one can compute more accurate volatility estimators such as realized variance, bipower variation, and jump variation. These measures allow for a decomposition of volatility into continuous and jump components.

Jump Detection Incorporating jump processes could improve volatility modeling, especially during periods of market stress.

Hybrid Models Combining GARCH-type models with implied volatility could lead to improved forecasts. For example, one could include the VIX as an exogenous variable in a GARCH-X framework.

Machine Learning Approaches Modern approaches such as LSTM networks or random forests could capture nonlinear patterns not accounted for by classical models.

Multi-Asset Analysis Extending the analysis to multiple asset classes (equities, FX, commodities) would improve the robustness and generality of the results.